Cell Switching for Energy Saving in Heterogeneous Small Cell Networks

異質小型基地台網路節能開關

楊智元 Zhi-Yuan Yang
Advisor: Ray-Guang Cheng

Outline

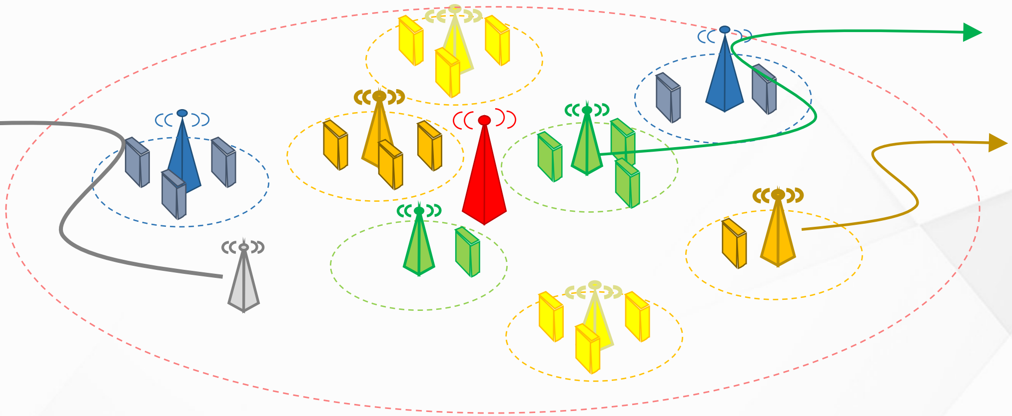
- Introduction
- System Architecture
- Proposed Method
- Experimental Results
- Conclusion

01 Introduction

Heterogeneous Small Cell Networks

Cell power model (active): $P(\rho[n]) = \underbrace{P^e}_{\text{Power efficiency}} * \underbrace{\rho[n]}_{\text{PRB usage}} + \underbrace{P^b}_{\text{Base power}}$ [1, 2, 3, 4]

Cell power model (inactive): P^{off} [2,4]

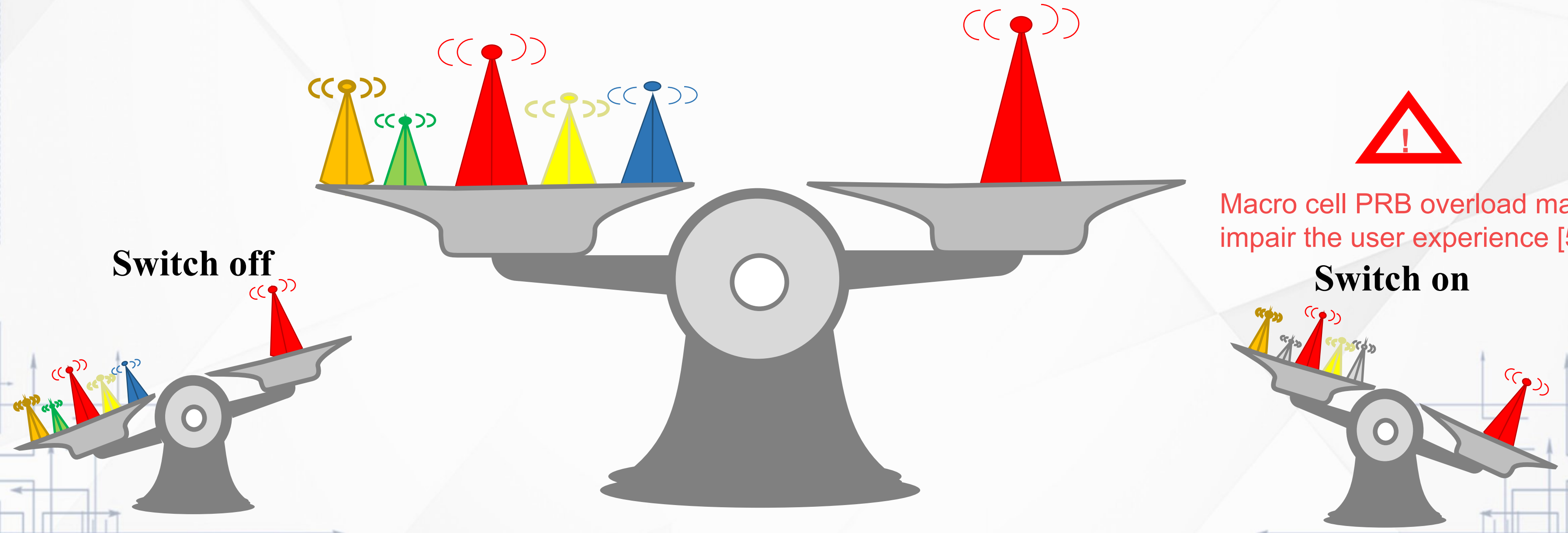


[1] O-RAN Global PlugFest Fall 2025 Energy Efficiency Testing: Venue COSMOS Joint North America O-RAN PlugFest (venue 08)
[2] X. Wang et al., "A Base Station Sleeping Strategy in Heterogeneous Cellular Networks Based on User Traffic Prediction," in IEEE Transactions on Green Communications and Networking, March 2024.
[3] J. Lin et al., "A Data-Driven Base Station Sleeping Strategy Based on Traffic Prediction," in IEEE Transactions on Network Science and Engineering, Nov.-Dec. 2024.
[4] K. Son et al., "Base Station Operation and User Association Mechanisms for Energy-Delay Tradeoffs in Green Cellular Networks," in IEEE Journal on Selected Areas in Communications, September 2011.

Power-load Tradeoff Cell Switching

Balance: **total power consumption** and **macro cell PRB usage**

Total power consumption **Macro cell PRB usage**



[5] O-RAN Alliance, "WG1 NESUC-R003-v02.00: Network Energy Savings Use Cases Technical Report," O-RAN Alliance, Tech. Rep., 2023. [Online]. Available: O RAN - Wg1.network Energy Savings Technical Report R003 v02.00 | PDF | Performance Indicator | Artificial Intelligence

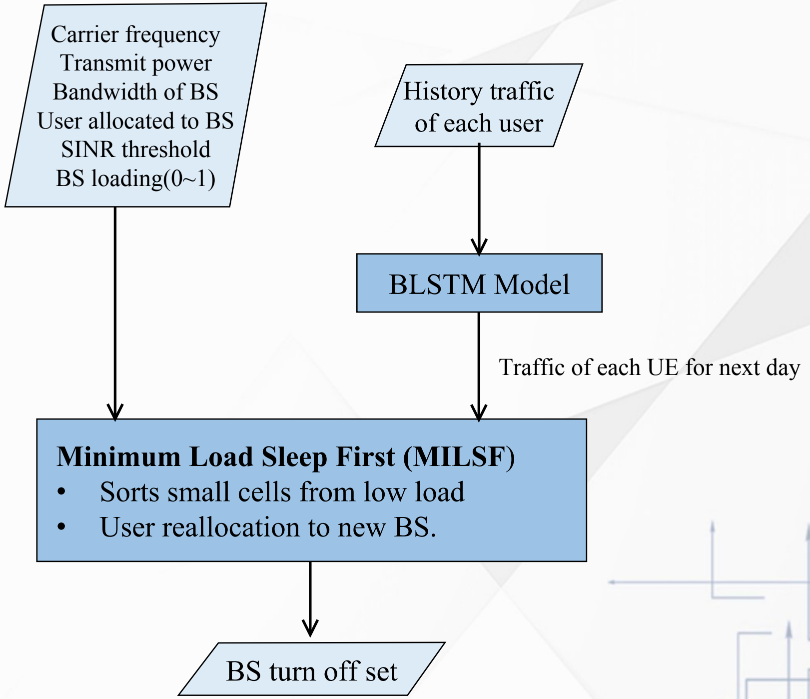
A Base Station Sleeping Strategy in Heterogeneous Cellular Networks Based on User Traffic Prediction *[2]*

Validation:

- **Dataset of UE**: Chinese cities with large populations.
- **Traffic prediction**: MAE and RMSE, throughput
- **Algorithm validation**: Energy saving percentage

Assumptions:

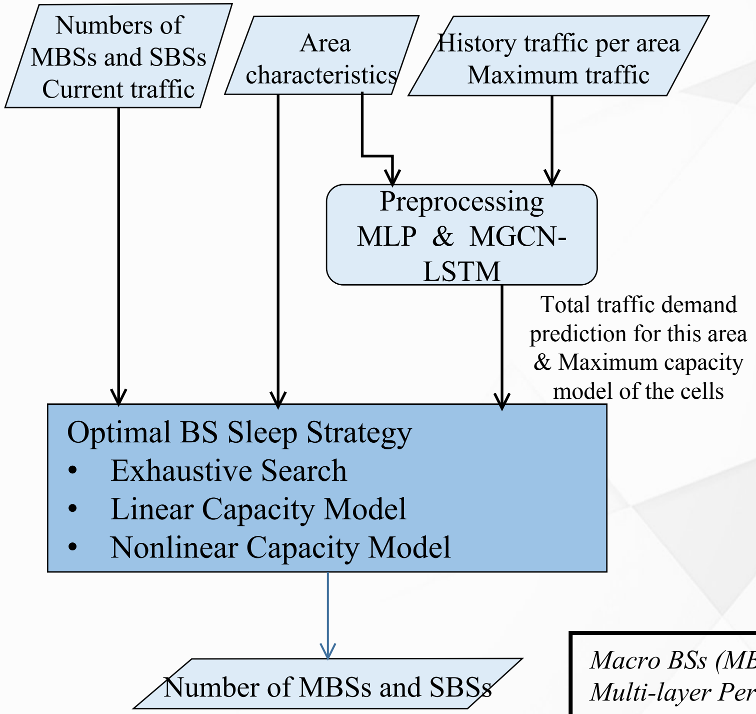
- Turn off the small cell only
- Operation during 10:00 p.m. and 6:00 a.m.



Bidirectional Long Short-Term Memory (BLSTM)
Base Station (BS)
Mean Absolute Error (MAE)
Root Mean Square Error (RMSE)

[2] X. Wang et al., "A Base Station Sleeping Strategy in Heterogeneous Cellular Networks Based on User Traffic Prediction," in IEEE Transactions on Green Communications and Networking, March 2024.

A Data-Driven Base Station Sleeping Strategy Based on Traffic Prediction *[3]*



Validation:

- **Dataset:** Real dataset of Milan**,** granularity: 1h
- **Traffic prediction**: MAE and RMSE
- **Algorithm validation**:
  - Energy saving percentage
  - Throughput and number of cells

Limitation:

- Homogeneous power consumption model of small cells

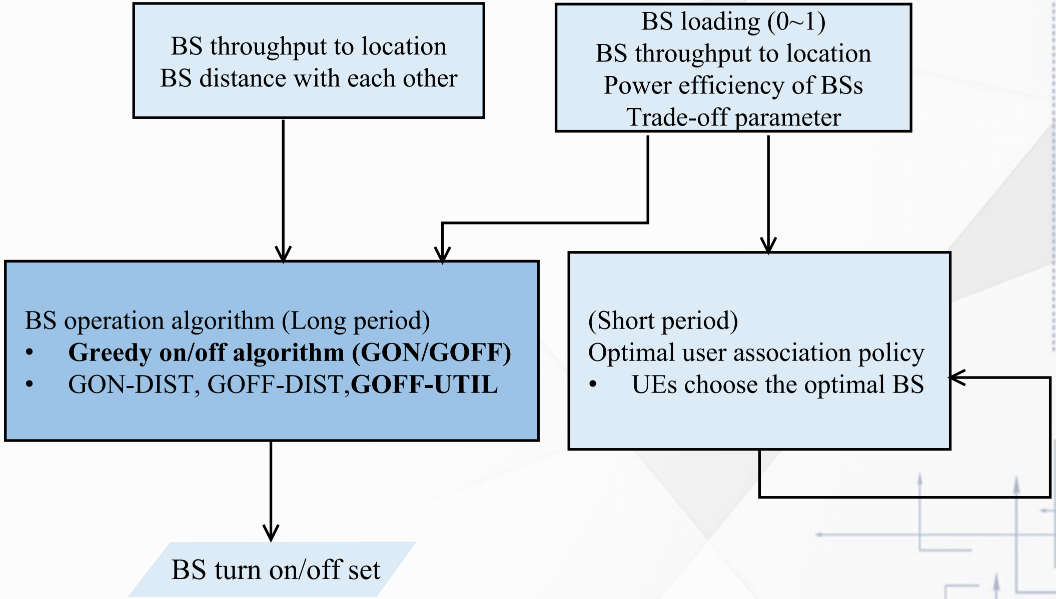
Macro BSs (MBSs) and Small BSs (SBSs)
Multi-layer Perceptron (MLP)
Multi-graph Convolutional Network (MGCN)

[3] J. Lin et al., "A Data-Driven Base Station Sleeping Strategy Based on Traffic Prediction," in IEEE Transactions on Network Science and Engineering, Nov.-Dec. 2024.

Base Station Operation and User Association Mechanisms for Energy-Delay Tradeoffs in Green Cellular Networks *[4]*

Validation:
- **Simulator:** Five macro BSs and five small BSs
- Algorithm validation: Compare different strategies
    1. Total power
    2. Energy Saving percentage vs. different traffic

Assumptions:
- Unlimited cell load and delay
- Inactive power is neglected



[4] K. Son et al., "Base Station Operation and User Association Mechanisms for Energy-Delay Tradeoffs in Green Cellular Networks," in IEEE Journal on Selected Areas in Communications, September 2011.

Related Work

Paper	Cell on/off	heterogeneous small cells	Inactive cell power	Algorithm	Evaluation Metrics
[2]	X / O	O	O	Turn off from Minimum Load	• Throughput and MAE and RMSE • Energy saving percentage
[3]	O / O	X	O	Determine the Minimum number of the macro and small cells	• MAE and RMSE • Throughput and number of active cells • Energy saving percentage
[4]	O / O	O	X	Turn on/off the benefit of delay reduction per power usages	• Total power consumption • Energy saving percentage
This Paper	O / O	O	O	Turn off from largest power saving	• Energy saving percentage • Throughput and macro cell overload percentage

[2] X. Wang et al., "A Base Station Sleeping Strategy in Heterogeneous Cellular Networks Based on User Traffic Prediction," in IEEE Transactions on Green Communications and Networking, March 2024.
[3] J. Lin et al., "A Data-Driven Base Station Sleeping Strategy Based on Traffic Prediction," in IEEE Transactions on Network Science and Engineering, Nov.-Dec. 2024.
[4] K. Son et al., "Base Station Operation and User Association Mechanisms for Energy-Delay Tradeoffs in Green Cellular Networks," in IEEE Journal on Selected Areas in Communications, September 2011.

Problem & Challenge & Contribution

Problem

- Reduce total power consumption in heterogeneous small cell network by switching on/off small cells without degrading users throughput requirement.

Challenge

- Cell switch off massive combination in heterogeneous small cells for energy saving.
- Trade-off between macro cell loading and energy saving.

Contribution

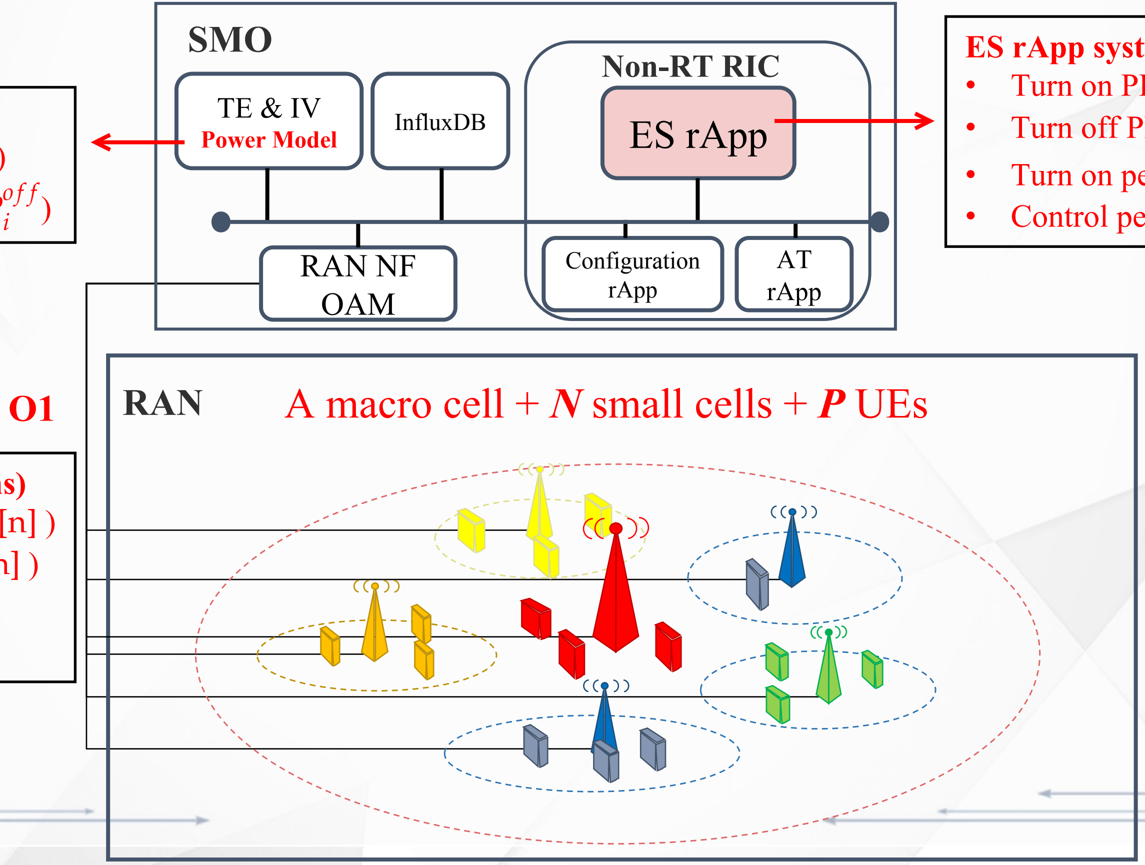
- A Max-Power-Saving cell turn off algorithm to optimize energy saving.
- Two threshold cell on/off strategy considering macro cell loading and energy saving.

02 System Architecture

System Architecture

Service Management and Orchestration (SMO)
Non-Real Time RAN Intelligent Controller (Non-RT RIC)
Topology Exposure & Inventory (TE & IV)
RAN Network Function Operation and Maintenance(RAN NF OAM)

Power model
- macro cell (P^e_M, P^b_M)
- small cell i (P^e_i, P^b_i, P^{off}_i)



ES rApp system parameter
- Turn on PRB threshold (μ_{on})
- Turn off PRB threshold (μ_{off})
- Turn on period (δ_{on})
- Control period (δ)

O1-PM: (Granularity: 10 mins)
- Macro cell PRB usage ($\rho^M[n]$)
- Small cell PRB usage ($\rho_i[n]$)

O1-CM
- Cell on/off (A)

Performance Metrics

Energy saving percentage: [2][3][4]

- $$\frac{\text{Total Energy without control} - \text{Total Energy with control}}{\text{Total Energy without control}} * 100\%$$

Average throughput: [2][3]

- $$\frac{\sum_{1}^{\text{Total simulation data point}} throughput_i}{\text{Total simulation data point}}$$

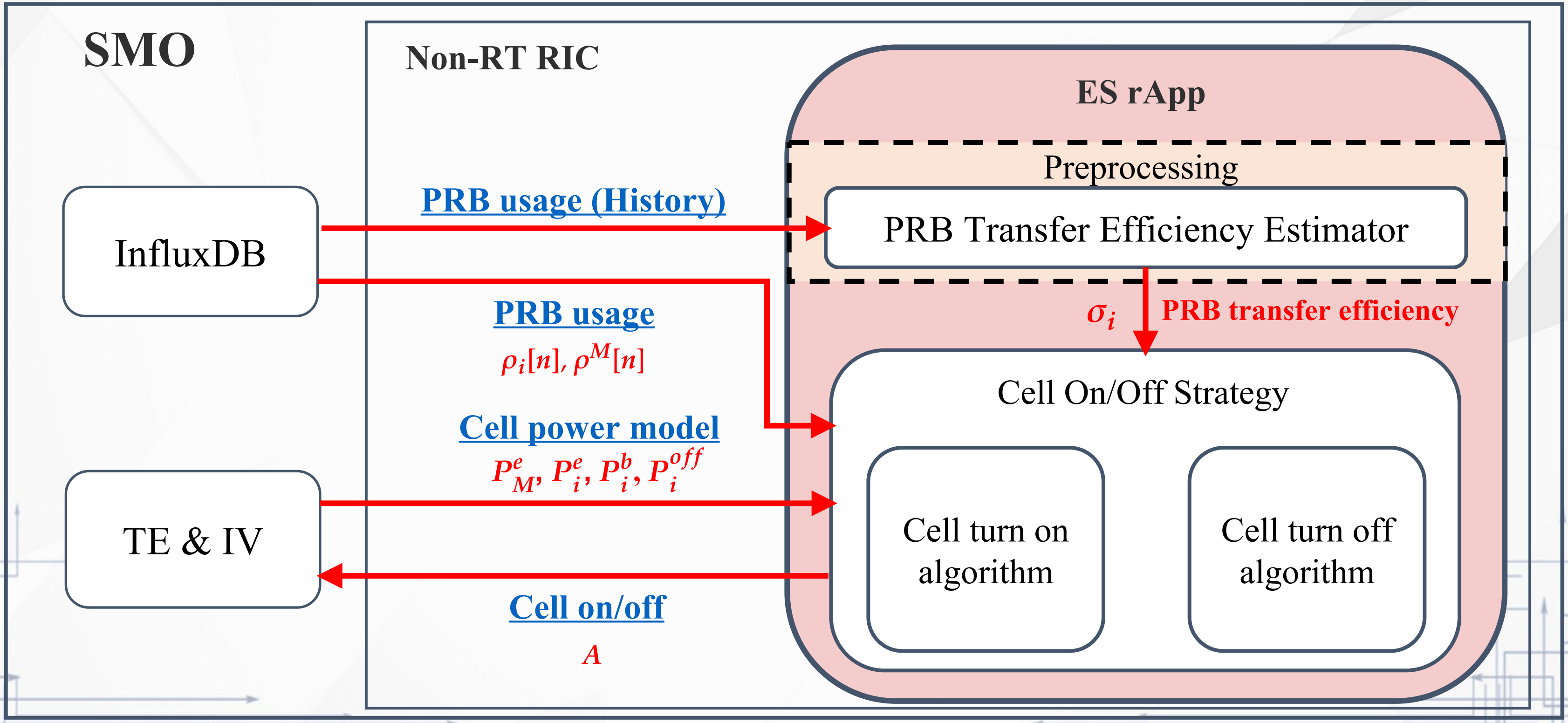
Macro cell full load percentage:

- $$\frac{\text{Number of full macro cell PRB data points}}{\text{Total simulation data point}} * 100\%$$

[2] X. Wang et al., "A Base Station Sleeping Strategy in Heterogeneous Cellular Networks Based on User Traffic Prediction," in IEEE Transactions on Green Communications and Networking, March 2024.
[3] J. Lin et al., "A Data-Driven Base Station Sleeping Strategy Based on Traffic Prediction," in IEEE Transactions on Network Science and Engineering, Nov.-Dec. 2024.
[4] K. Son et al., "Base Station Operation and User Association Mechanisms for Energy-Delay Tradeoffs in Green Cellular Networks," in IEEE Journal on Selected Areas in Communications, September 2011.

03 Proposed Method

Proposed Method



PRB Transfer Efficiency:

Macro Cell PRB Increment Estimation

Estimation:

- *UEs are uniform distribution in small cell i*
- *The average MCS drop after offloading is stable*

PRB Transfer Efficiency: $\sigma_i = \dfrac{R_i}{R_i^M} = \dfrac{\rho_i^M}{\rho_i}$

Average throughput per PRB for small cell ***i*** (Mbps/PRB)

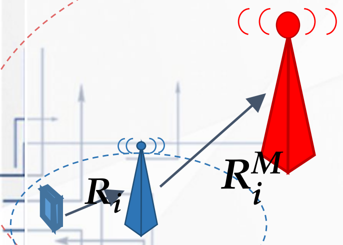
Throughput per PRB to macro cell under small cell ***i*** (Mbps/PRB)

$$T_i[n] = \rho_i[n] * R_i = \rho_i^M[n] * R_i^M$$

$$\rightarrow \rho_i^M = \rho_i * \frac{R_i}{R_i^M}$$

Macro cell PRB increment after closing small cell ***i*** (PRB)

Throughput of small cell ***i*** (Mbps)

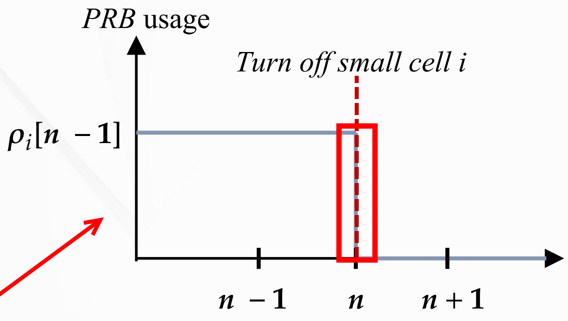


- *UEs are uniform distribution*

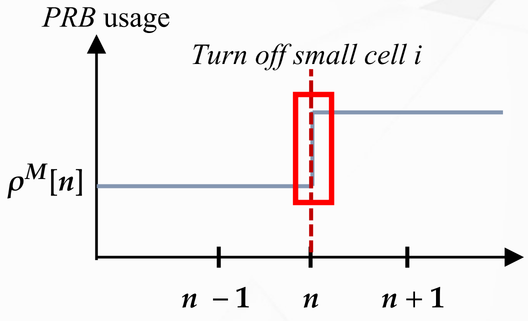
Modulation and Coding Scheme (MCS)

PRB Transfer Efficiency Estimator:
Modeling the PRB Transfer Efficiency

Small Cell



Macro Cell



History data

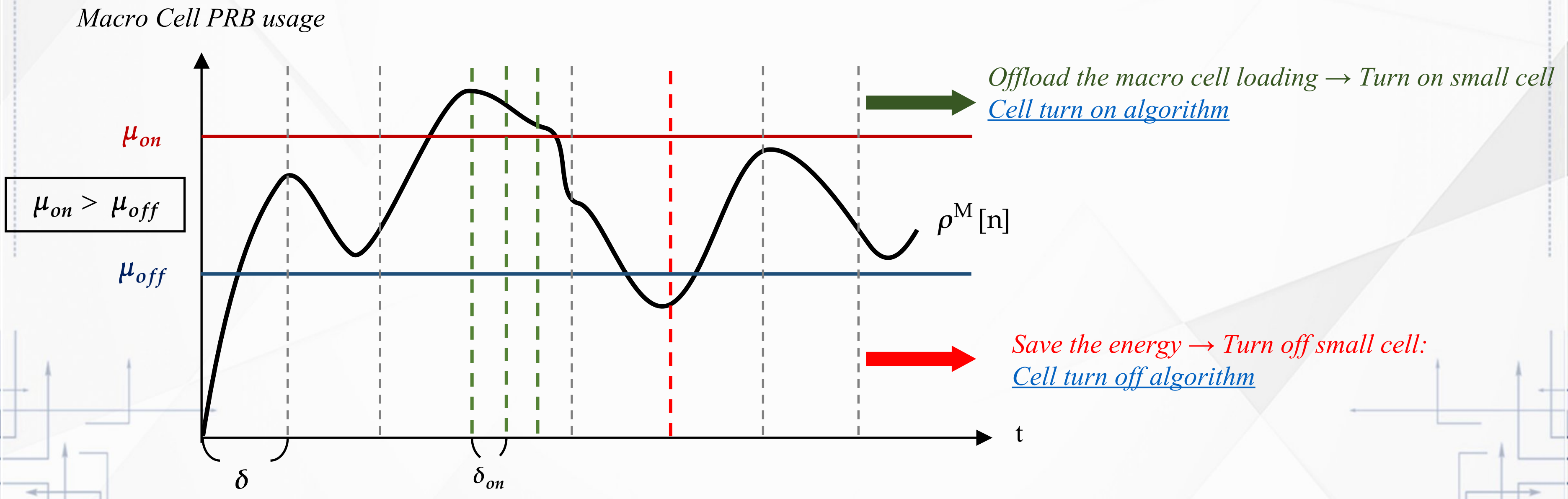
- PRB usage of **macro cell** before turn off cell i $(\rho^M[n-1])$
- PRB usage of **macro cell** after turn off cell i $(\rho^M[n+1])$
- PRB usage of **small cell i** before turn off $(\rho_i[n+1])$

Modeling the PRB Transfer Efficiency

$$\sigma_i = \frac{\rho^M[n+1] - \rho^M[n-1]}{\rho_i[n-1]}$$

σ_i

Cell On/Off Strategy



Cell Turn Off Algorithm

- **Goal:** Minimize total power consumption → **Maximize the power saving**
  - **Constraint:** Macro cell PRB usage($\rho^M[n]$) $< \mu_{off}$

- **Exhaustive Search**

Challenge: Combinatorial problem with $O(2^{|Cs|})$ possible cases [4]

$Cs \leftarrow$
Active small cells
$+\ P_i^{save}[n] > 0$

Cs

Find ***combination*** *of on/off each cell in Cs*

Choose maximum power saving
$+\ \hat{\rho}^M[n] < \mu_{off}$
combination

Power Saving by turning off cell i

Estimated Macro Cell PRB

[4] K. Son et al., "Base Station Operation and User Association Mechanisms for Energy-Delay Tradeoffs in Green Cellular Networks," in IEEE Journal on Selected Areas in Communications, September 2011.

Cell Turn Off Algorithm

Max-Power-Saving (MPS) algorithm

* *Combinatorial* $O(Nlog(N))$

Power model, PRB transfer efficiency, cell number of PRB and μ_{off}

1. $Cs \leftarrow$ ***Active small cells*** $+ P_i^{save}[n] > 0$
2. *Sort Cs by* $P_i^{save}[n]$: **large** to **small**

Iterative turn off cell i in Cs

$\hat{\rho}^M[n] \leq \mu_{off}$

Turn off set

- **Other Algorithm**

1. **Min-PRB-Increment (MPI):**

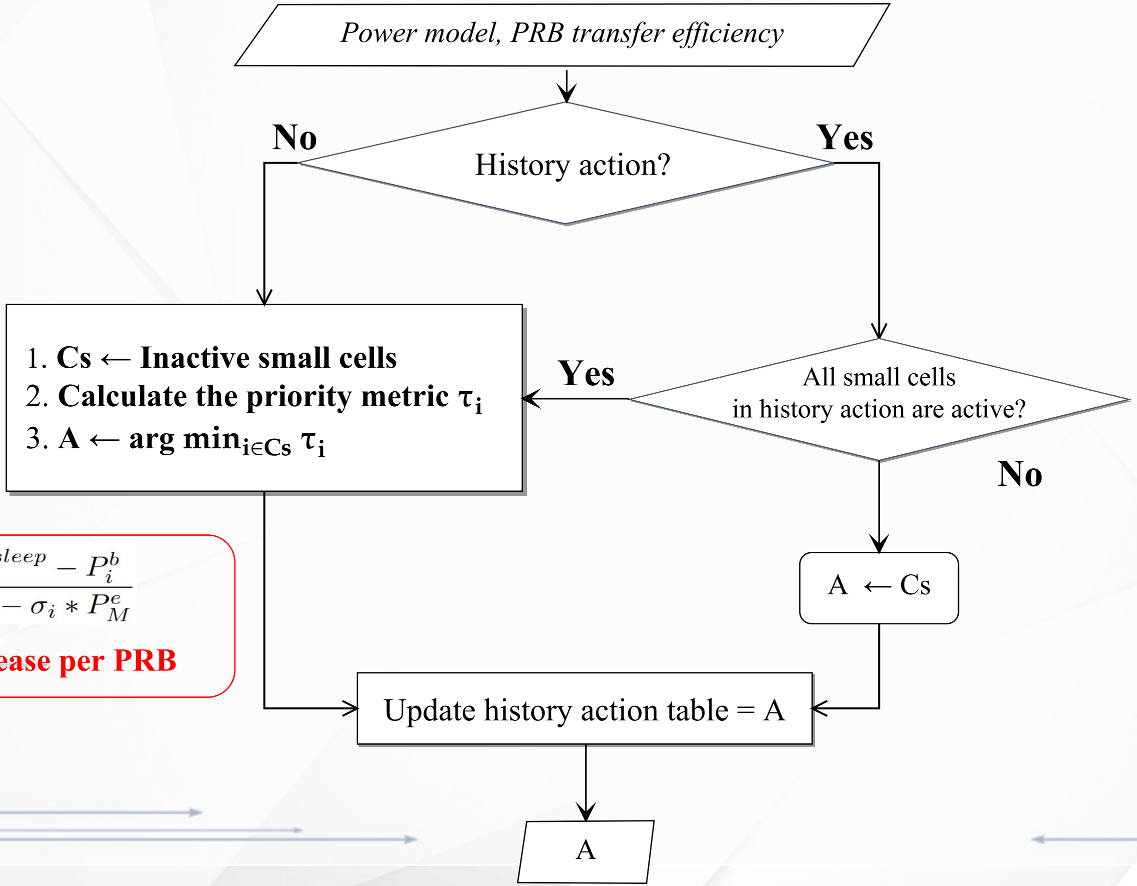
   1. $Cs \leftarrow$ ***Active small cells*** $+ P_i^{save}[n] > 0$
   2. *Sort Cs by* $\rho_i^M[n]$: **small** to **large**

   PRB usage offload to macro cell

2. **Min-PRB-Load (MPL):** *[2][4]*

   1. $Cs \leftarrow$ ***Active small cells***
   2. *Sort Cs form* ***small*** *to* ***large*** $\rho_i[n]$

Cell Turn On Algorithm



δ	
2δ	*C10, C11, C14*
...	
$n\delta$	

History action table

$$\tau_i = \frac{P_i^{sleep} - P_i^{b}}{P_i^{e} - \sigma_i * P_M^{e}}$$

Power increase per PRB

04 Experimental Results

Experimental Scenarios

Scenario 1: Algorithm Behavior Validation

- 【Goal】Verify whether the proposed MPS algorithm turns off cells.
  - **Fix: Same Cells loading**
  - **Vary: Same/Different power model**

Scenario 2: Algorithm Comparison under Random Loading

- 【Goal】Compare different cell on/off algorithms under realistic traffic.
  - **Fix: Random Cells loading**
  - **Vary: Same/Different power model**

Algorithm:

- Exhaustive Search
- **MPS**
- MPI
- MPL

Scenario 3: Cell On/Off Strategy and Thresholds

- 【Goal】Evaluate performance with on/off thresholds.
  - **Fix: 2 days simulation**
  - **Vary:** μ_{on} **and** μ_{off}

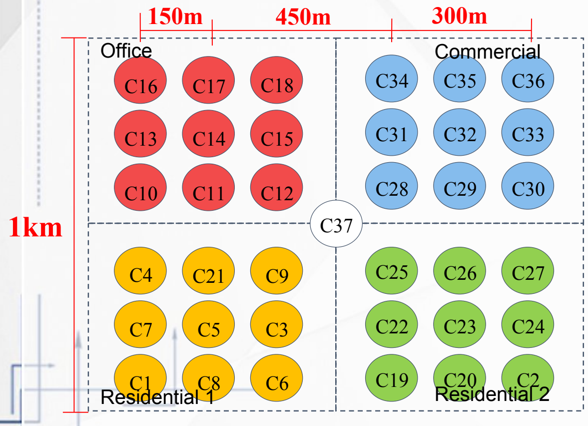
μ_{on}	**136 , 191, 245**
μ_{off}	**54 , 81, 109, 136, 163, 191, 218**

Experiment Configuration

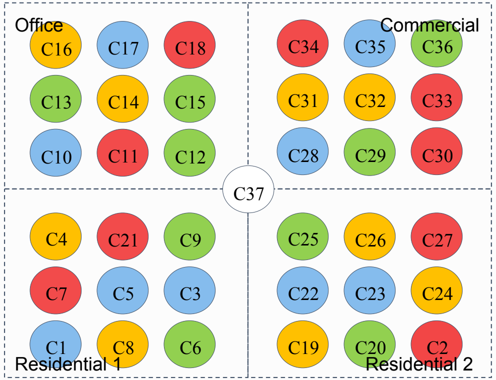
- RAN topology configuration in **VIAVI AI RSG**

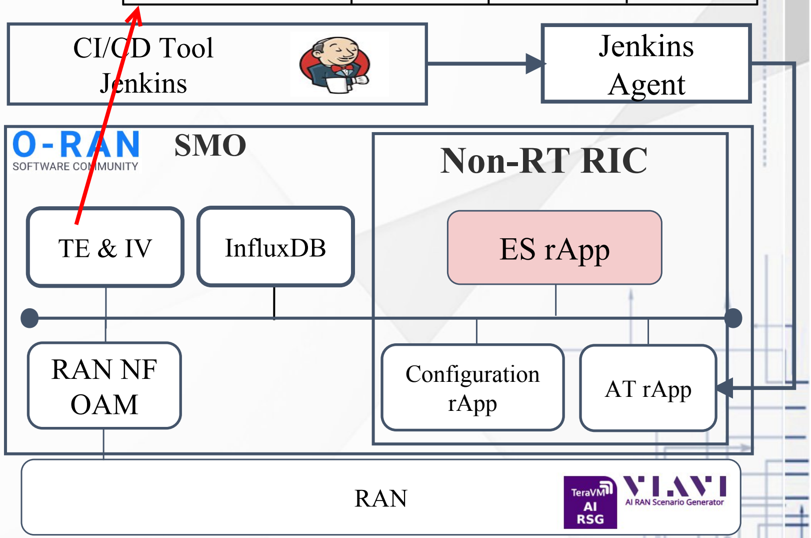
Cell	Power efficiency	Base power	Inactive power
Small cell type 1	0.1482	*10.5W*	
Small cell type 2	0.2024	*10.7W*	
Small cell type 3	0.2567	*10.9W*	*1W*
Small cell type 4	0.3290	*11.1W*	
Macro Cell	0.361	*51.25 W*	

Same power model



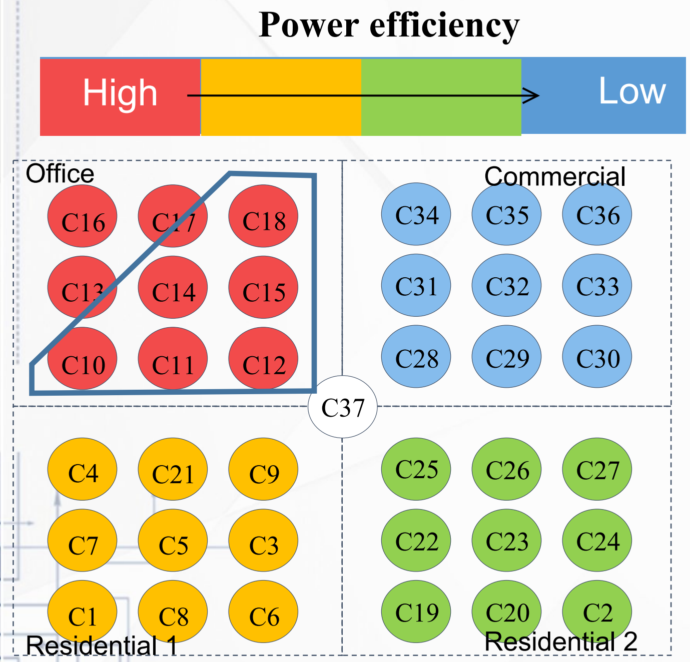
Different power model



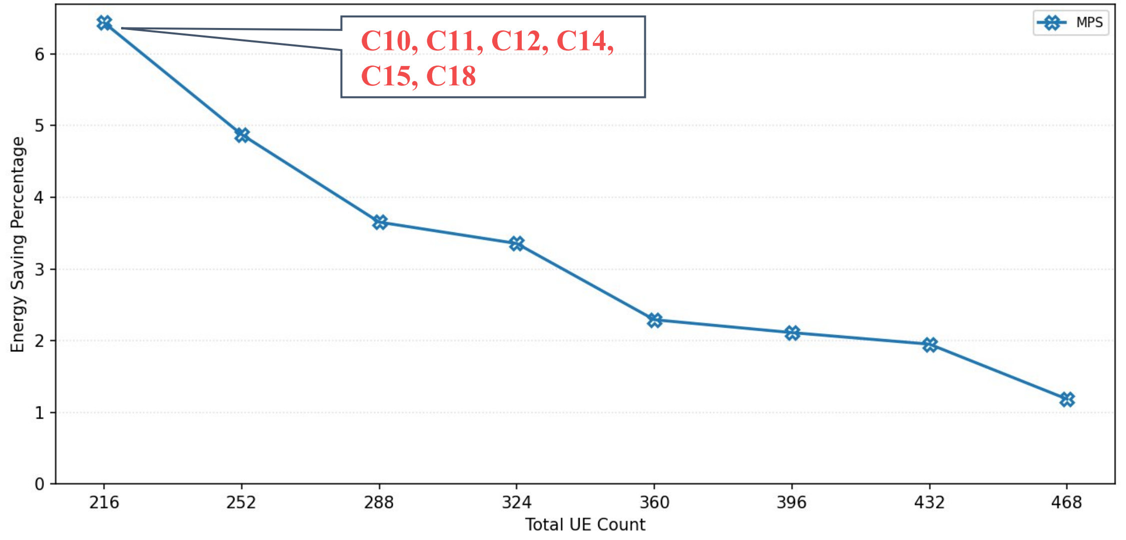


[7] EARTH Project, "Final report on green network architectures," Energy Aware Radio and Network Technologies (EARTH), Technical Report Deliverable D2.3, 2012.

Scenario 1: Algorithm Behavior Validation

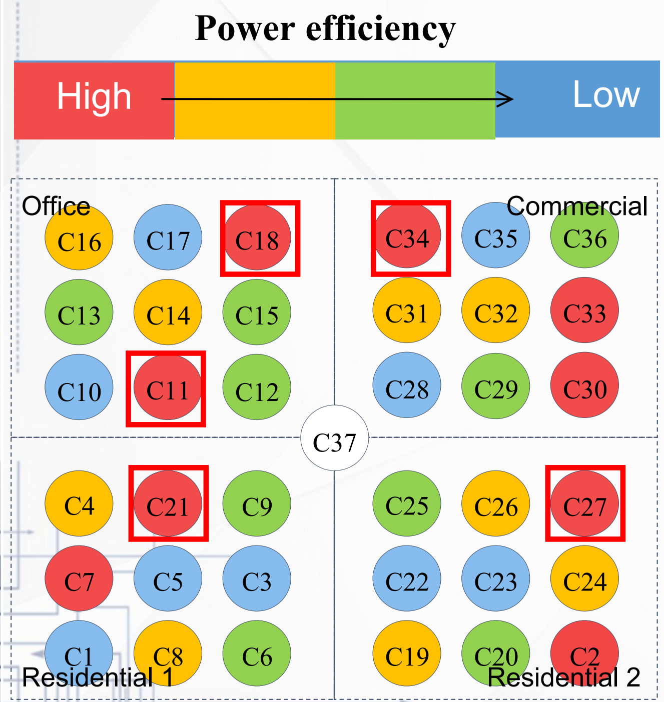


Choose closest macro cell and high power efficiency

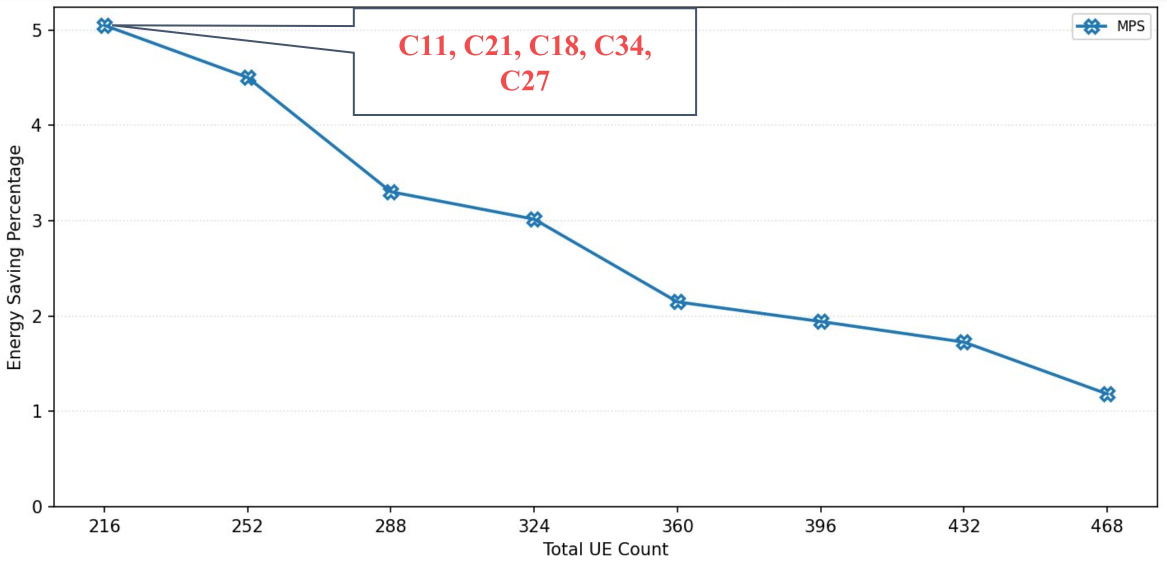


Number of UEs increase, energy saving percentage decrease

Scenario 1: Algorithm Behavior Validation



Choose **closest macro cell** and **high power efficiency**

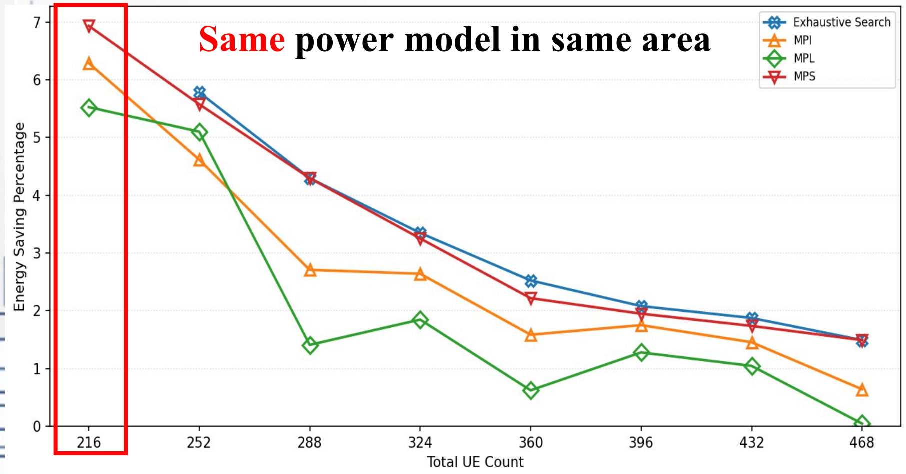


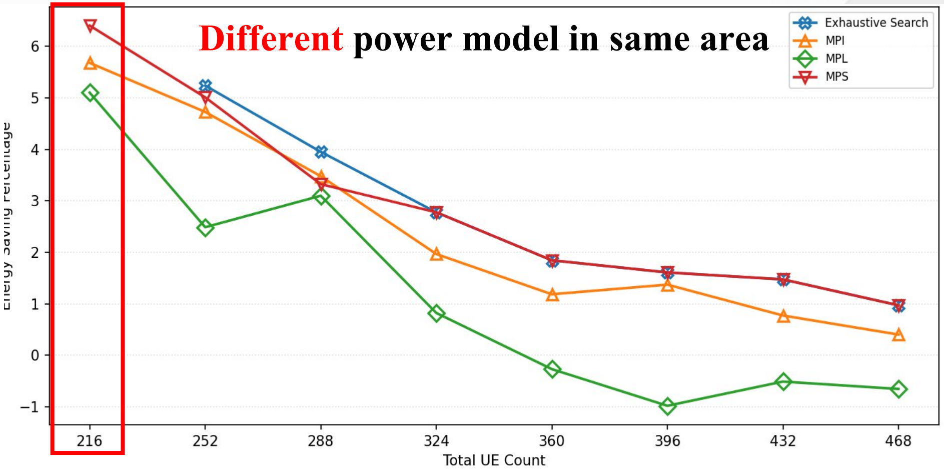
Scenario 2: Algorithm Comparison under Random Loading

✔ MPS performance closely approximates Exhaustive Search.

Random loading of each cell

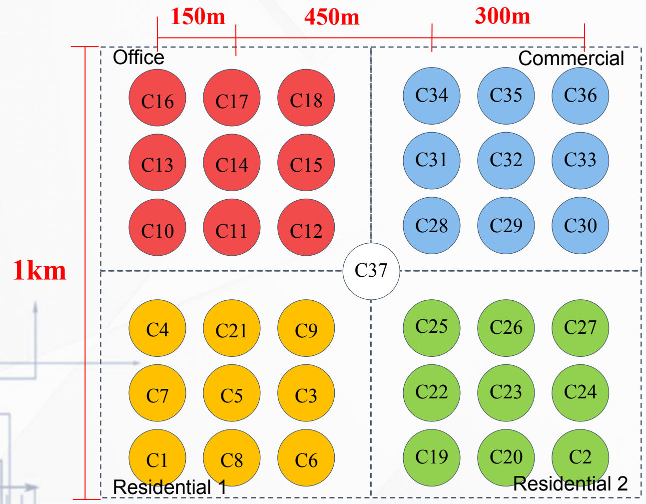
Exhaustive Search: Too many combination ($6.7 * 10^7$)



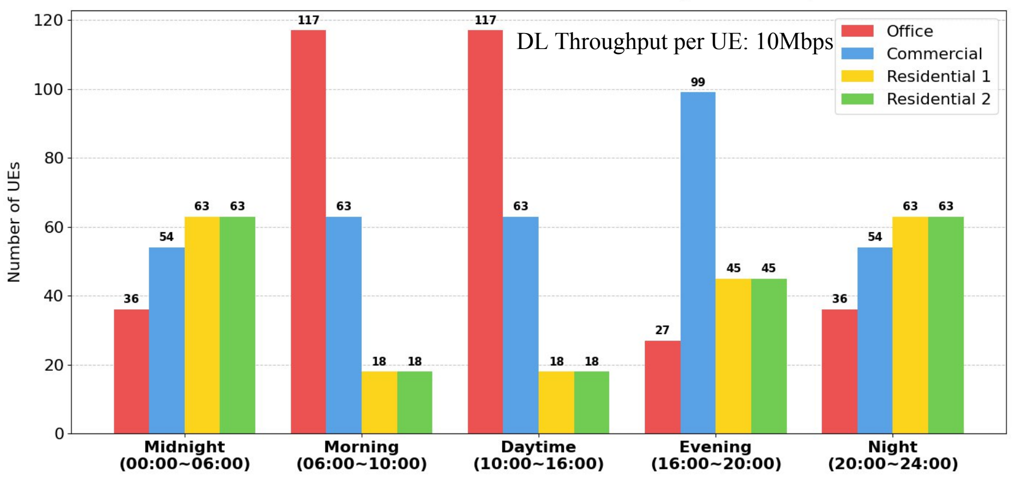


Scenario 3: Cell On/Off Strategy and Thresholds

- δ = 3600s, δon = 1200s
- Number of UEs: 216
- μ_{on}: **245, 191, 136** , μ_{off} : **54 , 81, 109, 136, 163, 191, 218**
- 2 days simulation.



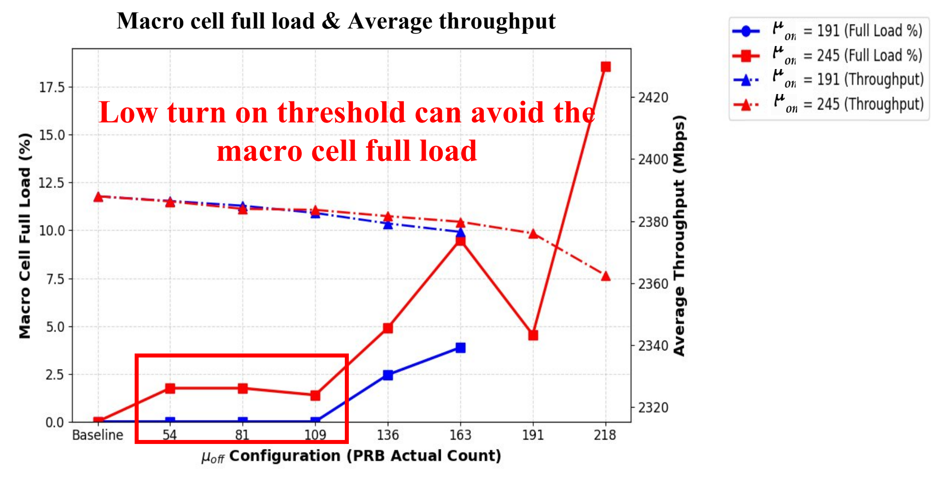
Number of UEs in each area with different time slot [6]

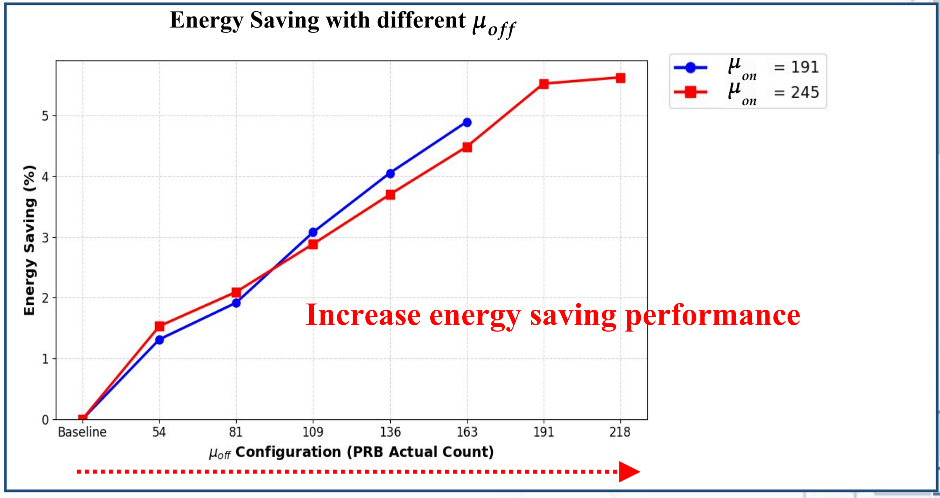


[6] Telecom Infra Project, "OpenRAN RIA Test Plan UC9: Power Saving by Load-Adaptive Mode," Telecom Infra Project (TIP), Test Plan, 2025, draft.

Scenario 3: Cell On/Off Strategy and Thresholds

μ_{on} : 245 vs. 191





Increase turn off threshold

Scenario 3: Cell On/Off Strategy and Thresholds

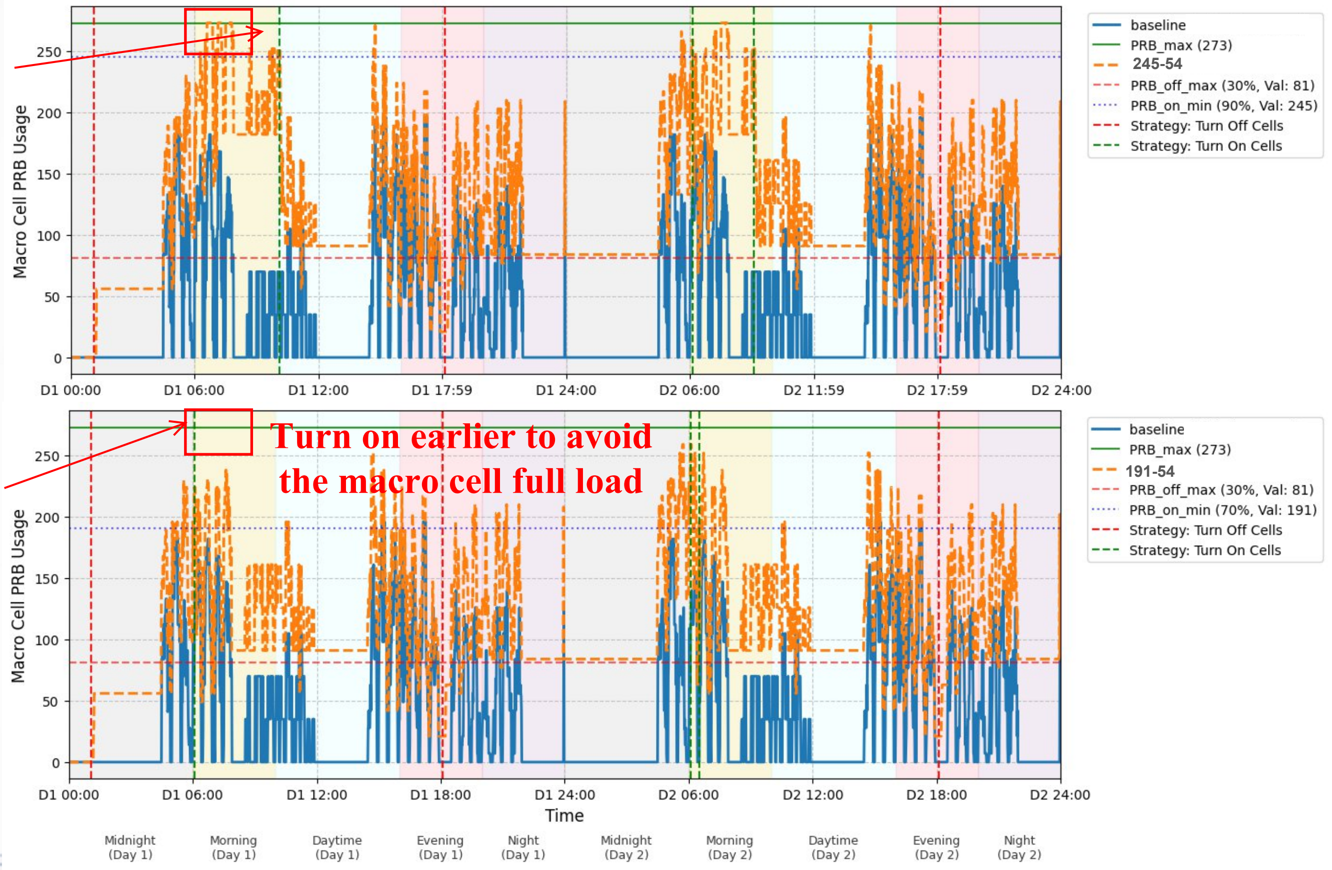
Turn on

μ_{on} : 245

μ_{off} : 81

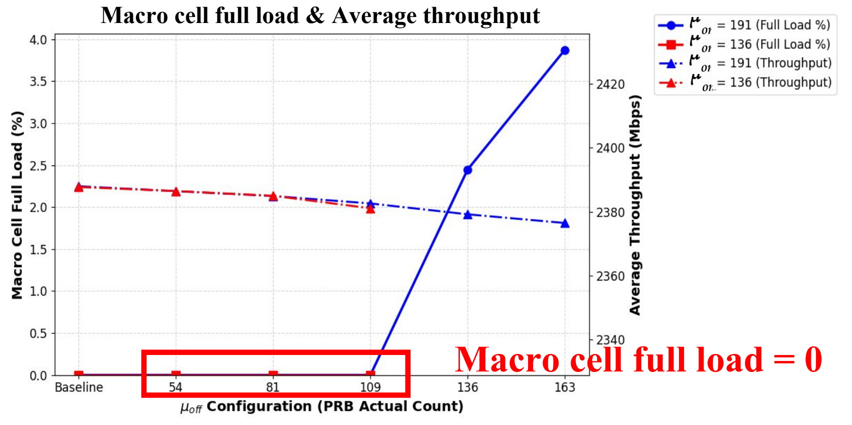
Turn on

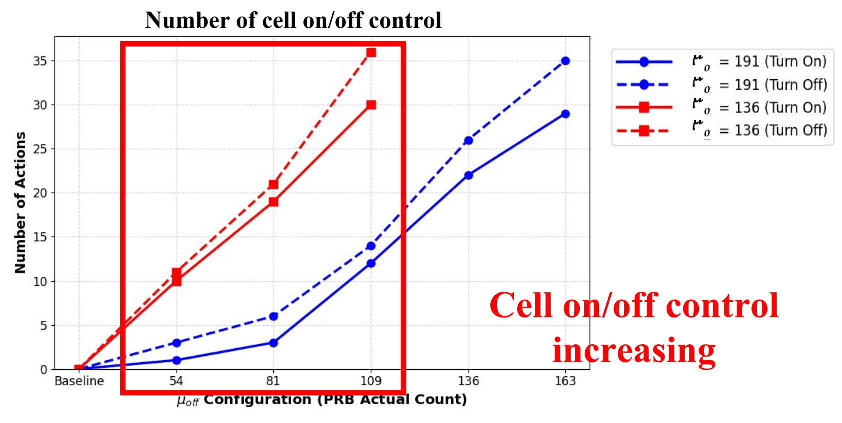
μ_{on}: 191

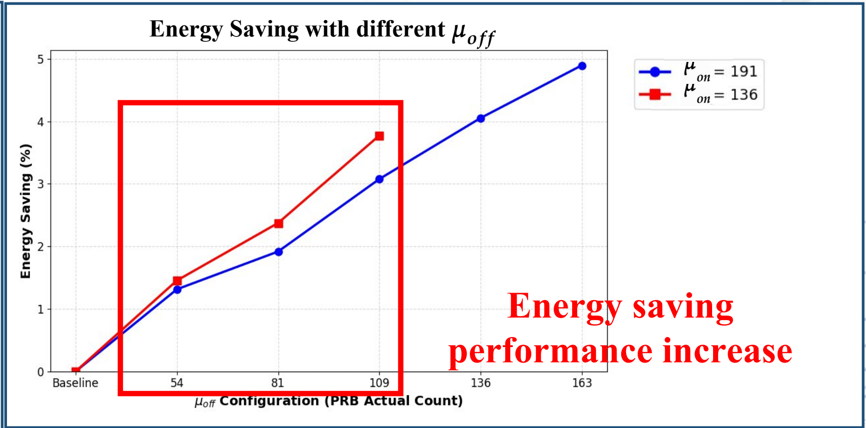


Scenario 3: Cell On/Off Strategy and Thresholds

Decrease μ_{on}: 191 → 136



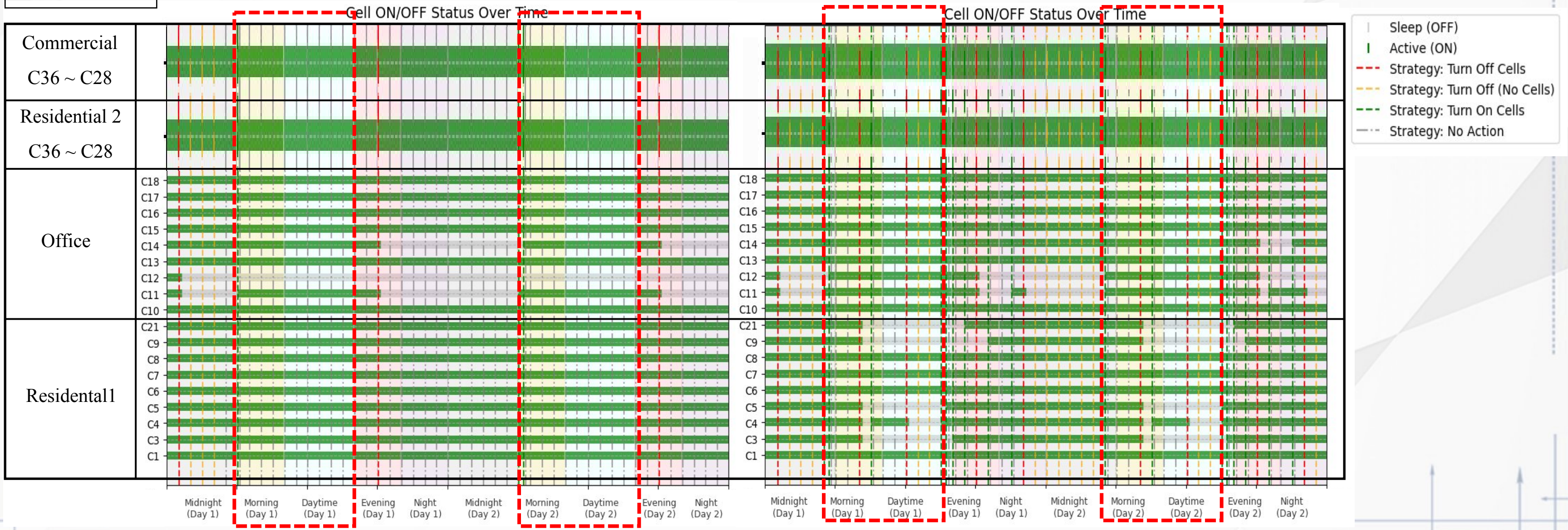




Scenario 3: Cell On/Off Strategy and Thresholds

μ_{off} : 81

μ_{on} : 191 μ_{on} : 136



Higher turn on threshold

Hard to turn off low loading cells

Lower turn on threshold

Switch to turn off low loading cells

05 | Conclusion

Conclusion

Conclusion

1. A Max-Power-Saving cell turn off algorithm to optimize power saving
   - Reduce the time complexity and energy saving result is close to Exhaustive Search
2. Propose a two threshold cell on/off strategy for cell capacity consideration and energy
   - Result shows the trade-off between the macro cell load and power saving

Future work

1. Locate the UEs location distribution through the UE measurement report for turn on.
2. Consider multiple macro cells scenario.

Demo video

https://youtu.be/H3k-hWMfMqc

Thank you